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The automation of leadership functions: Would people trust decision algorithms?[☆]Miriam Höddinghaus^{*}, Dominik Sondern, Guido Hertel

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ABSTRACT

The advancing maturity of algorithm-based decision-making enables computers to perform many leadership functions today. However, a central precondition of successful implementation should be that human workers trust such automated leadership agents. The present study ($N = 333$ workers) compared participants' reactions towards automated and human leadership using experimental vignette methodology with hypothetical work scenarios. We manipulated type of leadership agent (human vs. computer) and decision subject (disciplinary vs. mentoring), and measured participants' trustworthiness perceptions and trust in the leadership agent. Results showed that participants perceived automated leadership agents as being higher on integrity and transparency than human leadership agents. However, human leadership agents were perceived as more adaptable and more benevolent. No differences occurred with respect to perceived data processing capacity or as a function of decision subject. Perceived trustworthiness predicted trust in the leadership agent, which in turn was positively related to further work-related outcomes (e.g., perceived fairness of the decision, perceived organizational support), confirming the general relevance of trust for organizations. The results contribute to our understanding of trust in automated leadership and offer practical implications for computer-based decision-making in the leadership context.

1. Introduction

Companies no longer use digital information systems and algorithms for decision support only, but also for decision-making, resulting in increasing automation of many leadership functions (Chamorro-Premuzic & Ahmetoglu, 2016; Parry, Cohen, & Bhattacharya, 2016). For example, computers assign tasks to workers, evaluate the fulfilment of tasks, or determine compensation based on digital algorithms (Harms & Han, 2019). As a consequence, computers evolve from mere tools or information devices to leadership agents, creating a new power structure in human-computer interaction (e.g., Glikson & Woolley, 2020). The term “automated leadership” describes processes “whereby purposeful influence is exerted by a computer agent over human agents to guide, structure, and facilitate activities and relationships in a group or organization” (Wesche & Sonderegger, 2019, p. 200).

As automated leadership becomes more prevalent, we need to understand how people respond to this new quality of human-computer interaction (Lee, 2018). In particular, automated leadership should be

most accepted and successful when trusted by the “followers”. According to the integrative model of organizational trust (Mayer, Davis, & Schoorman, 1995), trust becomes particularly relevant in cases where the trustee (i.e., trust recipient) carries out an action relevant to the trustor (i.e., party who trusts) that includes certain risks for the trustor. This is quite often the case in hierarchical leadership contexts usually characterized by power asymmetries as leaders have the authority to make decisions that are very relevant for their followers (Dirks & Ferrin, 2002; Mayer et al., 1995). Moreover, automated leadership is associated with additional risks and uncertainties due to the lack of transparency and the high complexity of algorithms, which go beyond the dependencies and asymmetries in hierarchical leader-follower relationships between humans (Glikson & Woolley, 2020). Thus, we assume that trust in leadership contexts is essential, and should affect followers' willingness to accept and follow leadership decisions, which in turn could potentially facilitate further behavioral outcomes and attitudes relevant for the effectiveness of (automated) leadership.

So far, empirical research is scarce on trust reactions towards

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automated leadership. The current study addresses trust in automated leadership using an experimental vignette methodology with hypothetical work scenarios. Based on the integrative model of organizational trust (Mayer et al., 1995) as underlying theoretical framework, we examined workers' anticipated perceptions of trustworthiness as a function of agent type (automated vs. human leadership agent), their resulting trust in the leadership agent, and subsequent outcomes such as perceived fairness and intention to accept the decision. Moreover, given that trust in leadership might vary across different decision subjects, we considered decision type (disciplinary vs. mentoring) as potential moderator of the relationship between trustworthiness and trust.

Our study contributes to the emerging field of automated leadership in four ways. First, drawing upon conceptual work on trust in leadership and human-computer interaction, we conceptualize and examine trust processes with respect to automated leadership, responding to recent calls for more empirical research in this field (Harms & Han, 2019; Lee, 2018; Wesche & Sonderegger, 2019). Second, our study extends previous human-computer interaction research by considering a new power structure between humans and computers in which computers no longer have a subordinate or supporting role but adopt leadership functions. Third, we examine unique strengths and weaknesses of human and automated leadership agents, supporting a more reflected implementation of automated leadership. Fourth, the experimental approach of the current study allows causal inferences on the observed effects.

2. Theory and hypotheses

2.1. Automated leadership

The concept of automated leadership is inherent in various constructs within both the academic and the practitioner literature. Notions such as computer-human leadership (Wesche & Sonderegger, 2019), algorithmic leadership (Harms & Han, 2019) or management (Lee, Kusbit, Metsky, & Dabbish, 2015; Schildt, 2017), robot (-based) or robotic leadership (Samani & Cheok, 2011; Samani, Koh, Saadatani, & Polydorou, 2012) are used to describe human-computer interaction in which algorithms assume leadership functions. Lee (2018) defined such automated algorithms as “a computational formula that autonomously makes decisions based on statistical models or decision rules without explicit human intervention” (Lee, 2018, p. 18). For instance, algorithms are used to allocate tasks, set working paces (Wesche & Sonderegger, 2019), evaluate and optimize work outcomes across different industries (Lee et al., 2015), or to screen applicants and recruit talents (Langer, König, & Papathanasiou, 2019).

Automated leadership offers the potential to reduce the complexity of the leadership role as it provides additional resources for today's leaders by allowing them to delegate tasks and responsibilities. On a conceptual level, automated leadership corresponds with the idea of “substitutes for leadership” (Kerr & Jermier, 1978) because it offers task structure and performance incentives to such an extent that hierarchical human-human leadership is no longer necessary or influential (Howell, Bowen, Dorfman, Kerr, & Podsakoff, 1990). However, our previous understanding of both human-computer interaction and of leadership as human-human leadership is changing due to the shift in the power structure between humans and technology (Glikson & Woolley, 2020; Wesche & Sonderegger, 2019). Thus, we do not consider automated leadership as a mere substitute for leadership (i.e., comparable to task or subordinate attributes such as task routine or experience), but rather as a complement to human-human leadership that should be incorporated into a new, more comprehensive conceptualization of leadership.

Automated leadership offers a range of advantages, including decision speed, higher processing capacity, and cost reduction (Brynjolfsson & Mitchell, 2017; Glikson & Woolley, 2020). These benefits suggest that

algorithms might be better leadership agents than humans. However, automated leadership can pose particularly risky and uncertain contexts for workers. For example, the leadership agent in automated leadership processes is a computer, and thus quite different (even “alien”) to human workers. Particularly, the logic and complexity behind algorithm-based decisions rarely allows for complete transparency and is often not well understood (Ananny & Crawford, 2018; Glikson & Woolley, 2020). Indeed, only ten percent of participants in a recent survey (with a sample representative for Germany, $n = 1221$) indicated that they had already heard of algorithms and knew quite well how they worked (Fischer & Petersen, 2018). Moreover, workers usually have limited experience and knowledge of automated leadership as they are neither the person who developed the underlying algorithm nor a computer specialist. Additionally, the amount of data processed by an algorithm exceeds the capacity of a human being (Ferràs-Hernández, 2018). Accordingly, automated leadership can entail special risks and uncertainties for workers. As a result, trust, considered as the “willingness to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor, irrespective of the ability to monitor or control that other party” (Mayer et al., 1995, p. 712), should become particularly relevant. More specifically, trust should increase the likelihood that individual workers rely on automated leadership.

2.2. Leadership needs trust

In classical leadership research, which has so far largely focused on human-human leadership, trust is indeed considered a core requirement for effective leadership already (e.g., Colquitt, Scott, & LePine, 2007; Dirks & Ferrin, 2002). This is because trust plays a key role in initiating, establishing, and maintaining relationships (Balliet & Van Lange, 2013). Such social interaction processes are an inherent part of leadership, during which agents intentionally exert influence over other persons to guide, structure, and facilitate relationships and activities in organizations (e.g., Yukl, 2013). Additionally, leader-follower relationships are hierarchical in nature, giving supervisors authority and power. Leadership decisions have a major impact on workers (e.g., promotions, payment, layoffs), creating dependencies and uncertainties that also make trust essential in leader-follower-relationships (Dirks & Ferrin, 2002). Previous empirical findings support the importance of trust in the context of leadership, showing that trust in leadership relates to a variety of important work-related outcomes, such as job performance, organizational citizenship behavior, job satisfaction, and organizational commitment (e.g., Colquitt et al., 2007; Dirks & Ferrin, 2002).

Considering the importance of trust in leader-follower-relationships, it is essential to understand how trust evolves. In their integrative model of organizational trust, Mayer et al. (1995) suggested that trust emerges as a function of the trustor's general disposition to trust, and the extent to which the trustor perceives the trustee as trustworthy. Trustworthiness includes perceived ability, benevolence, and integrity. Ability refers to the set of skills, competencies, and characteristics that enable someone to exert influence in a particular domain. Benevolence is defined as the belief that the trustee will want to do good to the trustor and cares about its interests and feelings (Mayer et al., 1995). Further, empathy (i.e., the ability to sense other people's emotions) is an important aspect of benevolence (Bhattacharjee, 2002). Integrity refers to the belief that the “trustee adheres to a set of principles that the trustor finds acceptable” (Mayer et al., 1995, p. 719). Consistent and predictable behavior is an important factor in determining how workers assess the integrity of their supervisors. Finally, we consider transparency as a fourth trustworthiness component as previous research has identified it as another important determinant of trust across a variety of contexts (e.g., Breuer, Hüffmeier, Hibben, & Hertel, 2019; Glikson & Woolley, 2020; Pirson & Malhotra,

2011). Transparency describes behaviors and attributes that foster the understanding between trustor and trustee and contributes to the traceability of, for example, a decision (Breuer et al., 2019).

2.3. Automated leadership and trust

Given the significance of trust as an important mechanism for human-human leadership (Dirks & Ferrin, 2002; Mayer et al., 1995) and the heightened uncertainty embedded in automated contexts (Glikson & Woolley, 2020), trust should play a key role in the acceptance and adoption of automated leadership. To date, however, there are only few studies on trust in automated leadership. An exception is a study conducted by Lee (2018), which revealed that individuals had lower trust in automated leadership decisions than in human-made decisions when it comes to hiring decisions and performance evaluations. However, no trust differences occurred with respect to task assignments or scheduling. While these initial results are promising, both the determinants and the consequences of trust in automated leadership remain yet unclear.

Apart from this exception, research on trust in leadership has so far concentrated primarily on humans as leadership agents. Yet, the insights from human-human leadership are not easily transferred to automated leadership. In case of automated leadership, the trustee is a computer and therefore completely different from the usual human leadership agent, so that the trustee is less known and therefore more difficult to assess. Further, there are core differences between human and algorithmic leadership agents that may influence workers' perceptions and responses relevant for trust emergence (i.e., perception and assessment of trustworthiness). For instance, automated agents surpass humans in terms of decision speed and processing capacity (e.g., Brynjolfsson & Mitchell, 2017). At the same time, however, algorithms are not able to accurately recognize and respond to human emotions, and to show empathy to the same extent as humans (Chamorro-Premuzic & Ahmetoglu, 2016). Indeed, these differences between humans and technology have led to popular assumptions (e.g., MABA-HABA, Fitts List), suggesting that algorithms perform better for computational functions and humans perform better for social purposes (De Winter & Hancock, 2015; Glikson & Woolley, 2020).

In addition to extant research on trust in leadership, previous research on trust in technology has provided initial and very useful insights that may contribute to a better understanding of automated leadership as well. For example, empirical research on trust in technology has shown that technology trust correlates with the use of a specific technology (e.g., Lee & Moray, 1992; Lee & See, 2004), and even cognitive benefits (Hertel et al., 2019). In addition, a recent review summarizing research on trust in artificial intelligence (Glikson & Woolley, 2020) has demonstrated that properties such as tangibility, transparency, reliability, immediacy behavior, and anthropomorphism play a role in the emergence of trust in artificial intelligence.

However, the generalization of these findings to automated leadership is limited as well because previous research has largely addressed technologies as tools or interactive partners. Thus, the changed power structure between humans and computers inherent in automated leadership has not been considered thus far. With automated leadership, computers have control over human followers, creating feelings of dependence and uncertainty. Such feelings affect the development of trust, since attributing high trustworthiness is a particularly effective strategy to counteract the uncertainty caused by dependence and authority (Colquitt et al., 2007; Schilke, Reimann, & Cook, 2015).

To conclude, findings from both research on trust in (human) leadership and research on trust in technology cannot easily be transferred to

automated leadership. However, it stands out that the two research streams share very similar core assumptions about the development of trust. For instance, the models on trust in technology (McKnight, Carter, Thatcher, & Clay, 2011) or trust in automation (Muir, 1994) postulate equivalents of ability, benevolence, integrity (Mayer et al., 1995), and transparency (Breuer et al., 2019) to predict perceived trustworthiness of technologies. This structural coherence suggests that the factors influencing trust in humans and trust in technologies are rather similar in nature (e.g., Wang & Benbasat, 2005). Further, this corresponds to the "computers are social actors" (CASA) paradigm. Empirical studies based on this paradigm showed that people interacting with computers engage in social behaviors, such as reciprocity or politeness, apply similar social categories to computers, such as gender stereotypes, and form group relationships (Nass, Fogg, & Moon, 1996; Nass & Moon, 2000; Nass, Steuer, & Tauber, 1994).

Thus, trust in technology models and the CASA paradigm suggest that trust in automated leadership should be based on similar psychological mechanisms as human-human leadership. Of course, this does not mean that the two agents are attributed similar qualities and certainly not to the same extent, but simply suggests that the same criteria (i.e., trustworthiness, disposition to trust) determine the emergence of trust. Accordingly, we used the integrative model of organizational trust as the underlying theoretical framework for our hypotheses (see Fig. 1 for our proposed theoretical model).

2.4. Perceived trustworthiness of automated vs. human leadership

We expect that people attribute different qualities to automated and human leadership agents based on the inherent characteristics of both leadership agents. More specifically, we assume differences in the trustworthiness assessments of automated and human leadership agents with respect to the trustworthiness factors specified in the integrative model of organizational trust and its extensions (Breuer et al., 2019; Mayer et al., 1995).

With regard to the trustworthiness factor ability, we consider two facets as relevant in the context of automated leadership: data processing capacity and adaptability to changing conditions. This division was necessary to accommodate the different qualities of the two leadership agents and to make the differences in these two facets salient. While humans' actual data processing capacity is rather low due to various limitations regarding information processing in the human brain (e.g., visual short-term memory; De Winter & Hancock, 2015; Eppler & Mengis, 2004; Marois & Ivanoff, 2005), they are able to understand common-sense situations and make intuitive decisions based on subjective or non-logical factors such as sensitivity and emotion (Jarrahi, 2018). In contrast, algorithms can process millions of data points in a very short time (Ferrás-Hernández, 2018; Jarrahi, 2018), but tend to overemphasize objective, calculable criteria and are unable to evaluate subjective, largely non-calculable criteria (Parry et al., 2016). The primary reason is that artificial intelligence or its underlying algorithms can only be trained with regularities already contained in the data, so that everything that is not contained in the data does not exist for the algorithm, making it hard for them to adapt to new and unknown situations. Accordingly, generalizability to new situations and stimuli is considered one of the greatest challenges and quality criteria of current artificial intelligence systems (e.g., Chollet, 2017).

We assume that these differences between the two leadership agents in terms of their respective data processing capacity and adaptability are so evident that people are aware of them and thus perceive these differences. This corresponds to the compensatory principle or the so-called Fitts List (Fitts, 1951), according to which the function

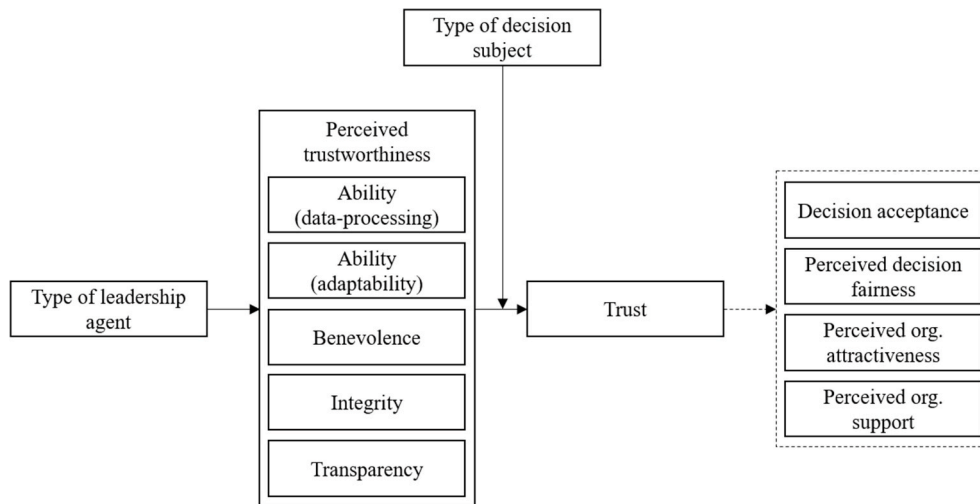


Fig. 1. Proposed theoretical model. Solid lines represents confirmatory research paths; dotted lines represent exploratory research paths.

allocation between humans and computers is based on their respective strengths and weaknesses. Under this principle, computers are perceived as superior in terms of qualities such as computing capacity or the handling of complex issues, whereas humans are perceived as superior in terms of their ability to improvise and use adaptive procedures. Recent empirical findings support this reasoning (De Winter & Hancock, 2015; Waytz & Norton, 2014). Accordingly, we assume¹:

Hypothesis 1a. The perceived data processing capacity of an automated leadership agent is higher than the perceived data processing capacity of a human leadership agent.

Hypothesis 1b. The perceived adaptability of an automated leadership agent is lower than the perceived adaptability of a human leadership agent.

Although the CASA paradigm (e.g., Nass & Moon, 2000) suggests that humans apply social rules to technology, it is unquestioned that computers have no emotions and are still unable to accurately detect and reciprocate human emotions (Chamorro-Premuzic & Ahmetoglu, 2016). Therefore, computers lack true human appreciation and empathy, which should be accentuated in interactions as they are unable to meet workers' needs for social interaction to the same extent as other humans. Benevolence by definition describes the extent to which a trustor believes that the trustee is willing to do him/her good (Mayer et al., 1995). Considering the lack of empathy, computers do not care about people, nor about their interests, so that the computer should be experienced as virtually neutral toward humans. As such, we assume:

Hypothesis 1c. The perceived benevolence of an automated leadership agent is lower than the perceived benevolence of a human leadership agent.

Human decisions can be influenced by a variety of cognitive distortions that lead to misjudgments (McShane, Nirenburg, & Jarrell, 2013). For example, humans tend to draw on prior experiences and beliefs, focus on limited interests and outcomes, rely on intuition rather than

logic or rational analysis (Korte, 2003), or use stereotypes (Koch, D'Mello, & Sackett, 2015). Given these numerous cognitive biases that influence human decision-making, algorithms might be attributed the potential to reduce biases in decision-making. They are known to make data-driven decisions based on predetermined, objective rules, so that the same inputs generate the same outputs each time (Chamorro-Premuzic, 2019; Lee, 2018; Merritt & Ilgen, 2008). In fact, research in human-computer interaction has already shown that humans perceive computers as generally consistent and reliable (e.g., Dzindolet, Peterson, Pomranky, Pierce, & Beck, 2003; Lee & See, 2004). Hence, we assume:

Hypothesis 1d. The perceived integrity of an automated leadership agent is higher than the perceived integrity of a human leadership agent.

Algorithms use complex techniques, statistics, and sometimes even machine learning (i.e., algorithms are programmed to train themselves) to make data-driven predictions (Brynjolfsson & Mitchell, 2017). Accordingly, automated decisions represent a “black box”—especially to people that have limited experience with and knowledge about algorithms (Glikson & Woolley, 2020). This is particularly true for algorithms that are continually improved and updated (Monteith & Glenn, 2016) as well as for machine learning or other types of artificial intelligence (e.g., neural networks). Certainly, the decisions of human leadership agents are often also not transparent and comprehensible for their followers. However, due to social categorization processes (e.g., Turner, 1985; Turner, Hogg, Oakes, Reicher, & Wetherell, 1987), human followers should perceive human leadership agents (same social category) as more similar than automated leadership agents (different social category). As a result, they should attribute more similar characteristics and behaviors (e.g., decision-making procedures) to humans that are, therefore, easier to understand (Ashforth & Mael, 1989; Stets & Burke, 2000). Hence, we assume:

Hypothesis 1e. The perceived transparency of an automated leadership agent is lower than the perceived transparency of a human leadership agent.

2.5. Perceived trustworthiness, trust, and the moderating effect of decision subject

Based on the integrative model of organizational trust (Mayer et al., 1995), we expect that the perception of an agent's trustworthiness is a direct antecedent of experienced trust. If a follower evaluates the agent's trustworthiness as favorable and accordingly perceives the agent as being able to fulfill leadership functions in an appropriate manner, the

¹ This study was pre-registered via the Open Science Framework (www.osf.io) to support open science (https://osf.io/qmtyu?view_only=01c946f42a0b4183b506166f21d66332). In the pre-registration, we developed individual hypotheses for each trustworthiness facet regarding its effect on trust, which we eventually combined into one Hypothesis (H2) for reasons of readability. Furthermore, we decided to treat the initial trust–outcome hypothesis as additional analyses to reduce the complexity of the study and analyzed our data using linear mixed models (LMMs) instead of ANOVA or SEM.

follower should be willing to rely on the leadership agent and to make him/herself vulnerable to the leadership agent (Mayer et al., 1995; Meeßen, Thielsch, & Hertel, 2019). This trustworthiness–trust relationship has already been validated both in the leadership (e.g., Colquitt et al., 2007) and in the human-computer interaction literature (e.g., McKnight et al., 2011; Wang & Benbasat, 2005). Hence, we assume:

Hypothesis 2. Perceived trustworthiness of the leadership agent predicts trust in the leadership agent.

Moreover, several researchers highlighted the role of contextual contingencies in the development of trust and stressed that the relevance of trustworthiness factors for trust emergence might differ across situations and contexts (e.g., Colquitt et al., 2007; Mayer et al., 1995; Pirson & Malhotra, 2011). A particularly important context variable in leadership relationships should be power (e.g., “having the discretion and the means to asymmetrically enforce one’s will over others”; Sturm & Antonakis, 2015, p. 139). Leadership situations can be understood as planned influence of some agents over other agents, with the result that power differences between trustor and trustee are inherent in most leader-follower relationships (Schilke et al., 2015).

Power differences give rise to feelings of dependence, which are often associated with increased risk perceptions due to e.g., a higher fear of exploitation (Colquitt et al., 2007; Schilke et al., 2015). Indeed, Sheppard and Sherman (1998) argued that the nature of dependence (i.e., power) influences the nature of risk and trust in relationships. Likewise, Pirson and Malhotra (2011) reported that the relevance of trustworthiness factors systematically varies between different stakeholders. However, there are different predictions regarding which facet of trustworthiness has a higher or lower relevance for trust emergence under which power structure. Some authors reported that it is only the integrity–trust relationship that should be stronger in situations involving high power differences. They argued that integrity related qualities, such as reliability, honesty, and fairness become even more important in cases where authority dynamics are particularly prominent (Colquitt & Rodell, 2011; Lind, 2001). Others suggested a high degree of dependence requires altruism, benevolence, and empathy in addition to integrity as these qualities also reduce the risk of cheating or exploitation (Sheppard & Sherman, 1998).

Given these differing views in the existing literature, we did not postulate a directed Hypothesis at the level of single trustworthiness factors. However, we did assume that power differences resulting from different decision subjects moderate the relationship between trustworthiness and trust. While there should be a large power difference in disciplinary decisions—and correspondingly higher uncertainties due to dependence and fear of exploitation—the power difference in mentoring decisions should be smaller. Accordingly, we assume:

Hypothesis 3. Decision subject moderates the relationship between perceived trustworthiness and trust in the leadership agent, such that the relative importance of the single trustworthiness factors changes depending on the decision subject.

2.6. Trust and workers’ reactions to leadership decisions

According to the integrative model of organizational trust, trust in leadership should influence workers’ attitudes and behaviors (Mayer et al., 1995). Specifically, trust should lead to more risk-taking behavior of the trustor, which in turn should influence workers’ intention to accept a leadership decision, perceived fairness of the decision, perceived organizational attractiveness, and perceived organizational support.

Furthermore, principles of social exchange (Blau, 1964) imply that workers who trust their supervisor are more likely to reciprocate the supervisor’s caring and considerate behavior (Gouldner, 1960; Jones & George, 1998), and thereby show positive work-related behaviors (Dirks & Ferrin, 2002). These general positive relations between trust and work-related behaviors and attitudes have already been supported by meta-analytical findings in various organizational contexts (Breuer, Hüffmeier, & Hertel, 2016; Colquitt et al., 2007; Dirks & Ferrin, 2002; Kong, Dirks, & Ferrin, 2014). Therefore, this connection is not the main focus of the present study. Nevertheless, in order to illustrate the relevance of trust in the context of the present study, we also measured a selection of outcome variables that are relevant in the context of leadership and decision-making: Intention to accept a leadership decision, perceived fairness of the decision, perceived organizational attractiveness, and perceived organizational support (extant theoretical and empirical work on these constructs are for example: Colquitt & Rodell, 2011; Langer, König, & Fritli, 2018; Meeßen et al., 2019; Tremblay, Cloutier, Simard, Chênevert, & Vandenberghe, 2010; Tyler, 1989, 1994).

3. Method

To test our hypotheses, we used an experimental vignette methodology with hypothetical scenarios. We chose this methodology because simulation-based procedures allow rather high experimental control due to the manipulation of constructs and are therefore a particularly suitable method to empirically approach an emerging research theme and provide a basis for the development and refinement of profound theory (Robinson & Clore, 2001). Participants were randomly assigned to a 2 x 2 (type of leadership agent: human vs. computer; type of decision subject: disciplinary decision vs. mentoring decision) mixed research design with type of leadership agent as within-subjects factor and type of decision subject as between-subjects factor. After participants were asked to imagine themselves in an application situation in which they have two similar job offers, two written vignettes were presented to the participants one after the other.

The vignettes were identical across all experimental conditions except for the parts containing experimental manipulations (see Appendix A for the full vignettes). Thus, the vignettes introduced either a human or an automated leadership agent as the party responsible for specific decisions in the context of the offered job position. Consequently, the presentation of the two leadership agents was very straightforward and clear, resulting in an evident manipulation of the first independent variable that was almost impossible for the participant to overlook. Further, we varied the decision subject (allocation of monthly bonus payments vs. allocation of trainings and workshops). We chose these decisions as exemplary leadership functions that can be automated and thus represent one specific type of automated leadership. Yet, we certainly cover only a part of potential leadership functions (i.e., our operationalization is just one way of approaching the concept of automated leadership). Furthermore, when selecting the decisions, we made sure to create realistic and immersive scenarios in order to meet the best practice recommendations for vignette studies by Aguinis and Bradley (2014). Both types of decision subjects reflect decisions that are likely to occur in work settings as they are relevant to everyday work and can potentially be made by algorithms. Additionally, we provided participants with contextual background information about the scenario and asked them to imagine themselves in the described situation as vivid and pictorial as possible. Thereby, we intended to improve the level of realism in our vignettes and to engage the participants to a greater extent.

3.1. Sample

Based on a power analysis with *G*Power* (Faul, Erdfelder, Lang, & Buchner, 2009) for a repeated measures ANOVA with within-between interaction (mostly equivalent to multi-level modelling), we approximated the sample size required for our analyses. 320 participants were necessary to have sufficient statistical power for detecting a small interaction effect ($\alpha = 0.05$, $1 - \beta = 0.80$). In total, 336 adults completed our study. We excluded three participants prior to analyses because they did not meet the inclusion criteria (i.e., consent given, sufficient German language skills). Thus, the final sample consisted of 333 German workers (206 females, $M_{age} = 48.55$, $SD = 10.03$). The majority of participants (62%) indicated a university degree as their highest educational level. Participants' industry affiliation was very diverse with the largest proportion of participants working in the service sector (14%), education and science (13%), technology and IT (11%), and health and medicine (10%). The remaining participants were from various industries such as public administration (8%), chemicals and pharmaceuticals (4%), and automotive (3%). All participants were recruited via the German online panel *PsyWeb* (<https://psyweb.uni-muenster.de/>). Members of this panel voluntarily agree to receive invitations for scientific surveys and are able to unsubscribe or delete their personal data at any time. For participation, participants received a feedback on their technology commitment, if wanted.

3.2. Procedure and manipulation

Participants received an email invitation via *PsyWeb* containing a link that provided access to the study. After following the link, participants read initial instructions and completed questionnaires assessing control variables. Afterwards, we introduced participants into the study scenario by a general description of the context (see Appendix A). We asked participants to imagine themselves in the final stage of an application process in which they have two concrete job offers hardly differing from each other.

Participants were randomly assigned to either the disciplinary decision ($n = 166$) or the mentoring decision condition ($n = 167$). Regardless of the decision subject, all participants first received the vignette describing the job position with a human leadership agent. Depending on the assigned condition, the decision referred either to the allocation of the monthly bonus payments (i.e., disciplinary decision) or the allocation of future trainings and workshops (i.e., mentoring decision). After reading the first vignette, we asked the participants to answer questionnaires measuring their reactions to the presented hypothetical scenarios including perceived trustworthiness of and trust in the leadership agent as well as the outcome variables. Afterwards, participants received the second vignette describing the job position with the automated leadership agent. The remaining procedure was similar to the first vignette. In the end, the participants completed a demographic questionnaire, followed by the debriefing. In total, it took participants about 20 minutes to conduct the experiment.

3.3. Measures

We adjusted items and instructions depending on the vignettes to address the leadership agent in the respective situation. Further, we translated items by using the common method of back-and-forth translation when no German version of a measure was available. Participants rated all items on a 7-point scale ranging from 1 (*strongly disagree*) to 7

(*strongly agree*). Appendix B provides a list with all items used for this study.

3.3.1. Dependent variables

All trustworthiness components were captured with three items. For ability (i.e., data processing and adaptability) and transparency, we developed items. Sample items are: "I believe that the supervisor has the competence to include all necessary information in the decision-making process" (data processing capacity; $\alpha_{human} = 0.93$, $\alpha_{automated} = 0.88$), "I think the computer can flexibly consider different circumstances when making decisions" (adaptability; $\alpha_{human} = 0.92$, $\alpha_{automated} = 0.92$), and "I think I could understand the decision-making processes of the computer very well" (transparency; $\alpha_{human} = 0.82$, $\alpha_{automated} = 0.90$). To measure benevolence and integrity we used items previously employed by Wang and Benbasat (2005) and adapted them to our study. A sample item for benevolence ($\alpha_{human} = 0.85$, $\alpha_{automated} = 0.90$) is "I believe that the supervisor would put my interests first". A sample item for integrity ($\alpha_{human} = 0.82$, $\alpha_{automated} = 0.72$) is "I believe that the computer makes unbiased decisions". We also averaged all 15 items to obtain an overall measure of perceived trustworthiness ($\alpha_{human} = 0.93$, $\alpha_{automated} = 0.85$). We measured trust using three items ($\alpha_{human} = 0.90$, $\alpha_{automated} = 0.92$) adapted from Thielsch, Meeßen, and Hertel (2018). A sample item is "I would heavily rely on the supervisor".

3.3.2. Additional variables

In addition to our dependent variables, we included a set of other measures for different reasons. First, we assessed four outcome variables to confirm the general trust–outcome link, which is part of the integrative model of organizational trust (Mayer et al., 1995). For perceived fairness of the decision, we developed a 3-item scale ($\alpha_{human} = 0.96$, $\alpha_{automated} = 0.90$) to ensure that the measure is aligned with the specific context of our study as suggested by Greenberg (1990). To measure participants' intention to accept the decision, we developed three items ($\alpha_{human} = 0.91$, $\alpha_{automated} = 0.93$). Further, we adapted three items ($\alpha_{human} = 0.93$, $\alpha_{automated} = 0.94$) of a scale developed by Highhouse, Lievens, and Sinar (2003) to assess organizational attractiveness. Finally, we measured perceived organizational support using four items ($\alpha_{human} = 0.86$, $\alpha_{automated} = 0.86$) of a scale developed by Kraimer and Wayne (2004).

Second, we included measures of dispositional trust to control for the influence of theoretically relevant individual dispositions that could serve as an alternative explanation for our findings (cf., Mayer et al., 1995). More specifically, we measured propensity to trust using five items ($\alpha = 0.89$) adapted from Ostendorf and Angleitner (2004) and general trust in technology using the 3-item scale ($\alpha = 0.92$) of McKnight et al. (2011). Further, we also included a measure of participants' risk perception (four items adapted from Jarvenpaa, Tractinsky, & Vitale, 2000) for possible exploratory analyses aimed at validating the entire integrative model of organizational trust (Mayer et al., 1995).

Finally, we measured work values (5 items adapted from Grube, 2009) and technology commitment (12 items by Neyer, Felber, & Gebhardt, 2012). While the technology commitment scale served as a reward for participation in our study, we assessed participants' work values to avoid a priming regarding the study content. Moreover, we asked the participants which of the two job positions, i.e., which leadership agent, they would choose if they had to decide between both offers ("If you now had to choose between the two positions, which job would you choose?").

Table 1
Descriptive statistics and intercorrelations among study variables.

Variable	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7	8	9	10	11	12	13	14
<i>Human agent</i>																
1. Trustworthiness	3.91	0.97	–													
2. Trust	3.28	1.39	.77*	–												
3. Fairness	3.90	1.20	.82*	.79*	–											
4. Acceptance	4.18	1.21	.65*	.67*	.64*	–										
5. Org. attractiveness	3.73	1.35	.58*	.55*	.56*	.61*	–									
6. Org. support	3.99	1.15	.62*	.58*	.61*	.55*	.72*	–								
<i>Automated agent</i>																
7. Trustworthiness	3.72	0.92	.29*	.21*	.23*	.25*	.27*	.24*	–							
8. Trust	3.09	1.57	.16*	.22*	.18*	.21*	.24*	.15	.67*	–						
9. Fairness	4.15	1.47	.20*	.14	.20*	.22*	.17*	.19*	.65*	.56*	–					
10. Acceptance	4.04	1.37	.20*	.23*	.16*	.41*	.27*	.25*	.61*	.68*	.61*	–				
11. Org. attractiveness	3.65	1.44	.21*	.19*	.18*	.28*	.56*	.40*	.52*	.58*	.42*	.58*	–			
12. Org. support	3.63	1.21	.36*	.30*	.34*	.35*	.49*	.66*	.51*	.40*	.39*	.50*	.65*	–		
<i>Controls</i>																
13. Propensity to trust	4.79	1.22	.36*	.33*	.32*	.16*	.14	.27*	.06	–.02	.03	–.00	–.02	.16*	–	
14. Trust in technology	4.42	1.46	.25*	.19*	.16*	.20*	.24*	.21*	.23*	.20*	.15*	.16*	.15	.19*	.28*	–

Note. *N* = 333. Org. = organizational.

*Holm-adjusted *p* < .05.

3.4. Data analysis

We analyzed our data using the statistics software *R* (version R 3.4.1). To test the hypotheses, we calculated multiple linear mixed models (LMMs) using the packages *lme4* (1.1–14) and *lmerTest* (3.1–0). We used LMMs to control for the statistical dependence of our data resulting from the repeated measurement of the dependent variables (i. e., two measurements are nested within one participant). Our experimental factors served as dummy-coded predictors in the LMMs. Human leadership agent and mentoring decision marked the reference group in the corresponding analyses. We initially controlled for dispositional trust and general trust in technology to eliminate theoretically relevant inter-individual differences that might affect our results. While both dispositional trust ($b = 0.18$, $SE(b) = 0.05$, 95%-CI [0.08; 0.28]) and general trust in technology ($b = 0.20$, $SE(b) = 0.04$, 95%-CI [0.11; 0.28]) significantly predicted trust, their inclusion into our analyses did not change the pattern of results. Thus, we report the analysis without control variables for purposes of simplicity. Across the analyses, the α -level was .05 and the reported *p*-values are two-tailed if not noted otherwise. Table 1 provides descriptive statistics and intercorrelations for our dependent and control variables.²

4. Results

The results of the linear mixed models (see Table 2) showed significant main effects for adaptability, benevolence, and integrity. We also found a significant main effect for transparency. However, the automated leadership agent was perceived as more transparent than the human leadership agent, which contradicts H1e. Further, we found no significant main effect for data processing capacity. Thus, H1b, H1c, and

² We conducted confirmatory factor analysis (CFA) to test construct validity of our main variables using the *R* package *lavaan* (0.5–23.1097). Results supported a six factor model for both the human and the automated leadership agent ($\chi^2_{\text{human}} = 340.52$, $df_{\text{human}} = 120$, $CFI_{\text{human}} = 0.95$, $RMSEA_{\text{human}} = 0.07$, $SRMR_{\text{human}} = 0.05$; $\chi^2_{\text{automated}} = 215.83$, $df_{\text{automated}} = 120$, $CFI_{\text{automated}} = 0.98$, $RMSEA_{\text{automated}} = 0.05$, $SRMR_{\text{automated}} = 0.05$) regarding our trust and trustworthiness variables. Various competing models were tested as well (i.e., one-factor, two-factor and five-factor models), but did not fit the data better than the six factor model. Thus, the CFA provided evidence for construct distinctiveness of the five trustworthiness components and trust in the leadership agent; and supported the six-factor structure of our constructs.

Table 2

Results of linear mixed model with trustworthiness factors as dependent variable and type of leadership agent (human vs. computer) as dummy-coded predictor.

Predictor	<i>b</i>	<i>SE(b)</i>	90% Confidence interval	
			<i>LL</i>	<i>UL</i>
Ability (data processing)				
Intercept	3.96***	0.08	3.83	4.10
Automated agent	0.03	0.10	−0.13	0.20
Ability (adaptability)				
Intercept	4.44***	0.08	4.31	4.56
Automated agent	−1.19***	0.10	−1.36	−1.02
Benevolence				
Intercept	3.17***	0.07	3.06	3.28
Automated agent	−1.30***	0.08	−1.43	−1.17
Integrity				
Intercept	3.93***	0.07	3.81	4.04
Automated agent	1.21***	0.09	1.07	1.36
Transparency				
Intercept	4.04***	0.08	3.92	4.16
Automated agent	0.31***	0.10	0.16	0.47

Note. The intercept denotes the mean of the reference group (human agent). *b* represents unstandardized regression weights. Regression weights denote differences between a group and the reference group. 90% confidence interval was used because of directed hypotheses. *LL* = lower limit of confidence interval; *UL* = upper limit of confidence interval.

****p* < .001, two-tailed using Satterthwaite's method.

H1d were supported, while H1a and H1e were not supported.³

The results of the linear mixed models conducted to test the effect of perceived trustworthiness on trust revealed a significant main effect (see Table 3). Thus, H2 was supported.⁴ As confirmatory factor analysis

³ Due to high correlations between the single trustworthiness factors, we also ran analyses in which we controlled for the remaining trustworthiness factors. Results slightly changed as the effect for data processing capacity got significant indicating that participants perceived the automated agent as having higher data processing capacity than the human agent ($b = 0.33$, $SE(b) = 0.12$, 95%-CI [0.13; 0.52]). Further, the effect for transparency did not reach the 0.05 α -level anymore ($b = -0.06$, $SE(b) = 0.12$, 95%-CI [–0.26; 0.14]).

⁴ In additional analyses, we tested whether type of leadership agent moderated the relationship between perceived trustworthiness and trust. We did not find a significant interaction effect.

Table 3

Results of linear mixed model with trust as dependent variable and trustworthiness as predictor.

Predictor	Main effect model				Interaction effect model			
	<i>b</i>	<i>SE(b)</i>	95% CI		<i>b</i>	<i>SE(b)</i>	95% CI	
			<i>LL</i>	<i>UL</i>			<i>LL</i>	<i>UL</i>
Intercept	3.01***	0.06	2.89	3.13	3.01***	0.06	2.89	3.13
Trustworthiness	1.14***	0.04	1.06	1.22	1.12***	0.06	1.01	1.23
Decision subject	0.35***	0.09	0.18	0.52	0.35***	0.09	0.18	0.52
Trustworthiness x decision subject					0.04	0.09	−0.12	0.21
Total <i>R</i> ²	0.53				0.53			

Note. The intercept indicates the expected level of trust when trustworthiness components are held constant at the grand mean. We used grand mean-centering to resolve multicollinearity issues (i.e., for the model with uncentered predictors, tolerance statistics indicated multicollinearity). *b* represents unstandardized regression weights. Regression weights denote the increase in trust for a change of one unit in the predictor holding the other predictors constant. Total *R*² = pseudo *R*² calculated with *R* package *r2glmm* (0.1.2) using the approach by Nakagawa, Johnson, and Schielzeth (2017); CI = confidence interval; *LL* = lower limit; *UL* = upper limit.

****p* < .001, two-tailed using Satterthwaite's method.

revealed a significantly better fit for the model containing the single trustworthiness factors, we also conducted the analysis with the six factor model. Results indicated the same pattern as there were significant positive main effects for all five trustworthiness factors with trust.

In addition to our confirmatory analyses regarding H1a–e and H2, we also explored differences between the two leadership agents with respect to the perceived overall trustworthiness as well as trust, and tested whether trustworthiness mediated the relationship between type of leadership agent and trust. Overall, participants perceived the human leadership agent (*M* = 3.91) as significantly more trustworthy than the automated leadership agent (*M* = 3.72, *b* = −0.19, *SE(b)* = 0.06, 95%-CI [−0.31; −0.06]). Also, the human leadership agent (*M* = 3.28) was ascribed a higher level of trust than the automated leadership agent (*M* = 3.09). However, this difference slightly failed to reach the .05 *α*-level (*b* = −0.19, *SE(b)* = 0.10, 95%-CI [−0.39; 0.01], *p* = .06), thus was not significant. To test the indirect effect between type of leadership agent and trust for significance, we calculated confidence intervals using level-2 bootstrapping. Analyses revealed a significant indirect effect (*ab* = −0.21, 95%-CI [−0.35; −0.07]) indicating a mediation-consistent pattern.

We found no significant interaction effect for overall trustworthiness and decision subject. However, analyses on the level of the five single trustworthiness components indicated a significant interaction effect for integrity and decision subject (*b* = 0.19, *SE(b)* = 0.07, 95%-CI [0.06; 0.32]). For disciplinary decisions, integrity perceptions had a stronger positive effect on trust as compared to mentoring decisions, probably reflecting stronger fairness concerns. Thus, H3 was only partially supported. Stepwise model comparisons confirmed the results (see Table 4).

In addition to our confirmatory analyses, we explored whether trust predicted a set of outcomes to demonstrate the relevance of trust in the context of automated leadership. Results (see Table 5) indicated that trust in the leadership agent was positively associated with all outcomes (i.e., perceived fairness of the decision, acceptance of the decision, organizational attractiveness, and perceived organizational support).

Table 4

Results of stepwise model comparisons.

Model	<i>df</i>	<i>AIC</i>	Deviance
Baseline (only random intercept)	3	2405.4	2399.4
+ Trustworthiness	4	1929.0	1921.0***
+ Decision subject	5	1915.1	1905.1***
+ Trustworthiness x Decision subject	6	1916.8	1904.8

Note. We entered the predictors in a stepwise manner to test every predictor for significance.

****p* < .001.

Table 5

Results of linear mixed model with outcomes as dependent variable and trust as predictor.

Predictor	<i>b</i>	<i>SE(b)</i>	95% Confidence interval	
			<i>LL</i>	<i>UL</i>
Fairness				
Intercept	2.16***	0.10	1.97	2.35
Trust	0.59***	0.03	0.53	0.64
Acceptance				
Intercept	2.31***	0.09	2.13	2.49
Trust	0.57***	0.02	0.52	0.61
Org. attractiveness				
Intercept	2.18***	0.10	1.98	2.38
Trust	0.47***	0.03	0.42	0.53
Org. support				
Intercept	2.77***	0.09	2.59	2.95
Trust	0.33***	0.02	0.28	0.37

Note. There is no meaningful interpretation of the intercepts due to the scaling of the predictor. *b* represents unstandardized regression weights. Regression weights denote the increase in the outcome for a change of one unit in trust. *LL* = lower limit of confidence interval; *UL* = upper limit of confidence interval; Org. = organizational.

****p* < .001, two-tailed using Satterthwaite's method.

5. Discussion

The present study represents one of the first studies investigating workers' responses towards automated leadership, thus addressing the changes in power and authority relations in human-computer interaction. We used an experimental vignette design with hypothetical scenarios to investigate the perceived trustworthiness of and trust in automated leadership agents as compared to human leadership agents. Thereby, our study responds to recent technological developments and calls for research that advances our understanding of variables affecting the perception and acceptance of automated leadership (e.g., Harms & Han, 2019; Wesche & Sonderegger, 2019).

Our results coincide with studies investigating trust in human-human leadership and trust in technology (as support), although we considered the new power structure between humans and computers in which computers are in a more hierarchical leadership position as trust recipient. More specifically, our findings suggest that trust in both human and automated leadership agents evolves as a function of their perceived trustworthiness. However, we also found that both agents have different strengths and weaknesses with respect to different factors of trustworthiness. While human leadership agents were perceived more benevolent and adaptable, participants attributed higher levels of integrity and transparency to the automated leadership agent. Interestingly, we found

no differences in the perceived data processing capacity. Combining all trustworthiness components into a single factor, results indicated that participants perceived the traditional form of leadership (i.e., the human leadership agent) as more trustworthy. With respect to decision subject as moderator of the trustworthiness–trust relationship, we found that decision subject moderated only the relationship between integrity and trust. Results indicated that the relationship was stronger for disciplinary decisions compared to mentoring decisions. Further, additional analyses supported trust as a predictor of the outcomes decision fairness and acceptance, and organizational attractiveness and support.

5.1. Theoretical contributions

The findings of our study indicate that trust in automated leadership can be described in terms of extant trust taxonomies from related literatures that conceptualize trust as a function of different trustworthiness components. Accordingly, it seems that trust in automated leadership agents is determined by the computers' perceived trustworthiness, suggesting that we can apply existing trust models (e.g., Mayer et al., 1995; McKnight et al., 2011; Muir, 1994) to automated leadership. Further, our findings support the key assumptions of the CASA paradigm as people perceive human characteristics in automated leadership (e.g., Nass & Moon, 2000).

Although the results of our study suggest that existing trust taxonomies generally apply to automated leadership, they also showed differences between the two forms of leadership. In line with our assumptions, automated and human leadership agents differed with respect to the attributed levels of ability, benevolence, integrity, and transparency. For instance, participants assessed human agents as more adaptive and benevolent. These findings are consistent with the view that humans are able to consider qualitative factors in their decisions, while computer-made decisions are entirely based on objective, predictable criteria, which might disregard the interests and emotions of workers (Parry et al., 2016). Thus, our results support the assumptions known from the MABA-HABA or Fitts list approach, suggesting that humans surpass algorithms in their ability to improvise and use adaptive processes (De Winter & Hancock, 2015; Fitts, 1951). With respect to integrity, participants ascribed higher levels to automated agents. This is in line with prior research showing that humans consider computers as reliable and consistent (e.g., Dzindolet et al., 2003; J. D.; Lee & See, 2004).

Contrary to our Hypothesis, participants perceived automated leadership agents as more transparent than human leadership agents. Given the complex statistical techniques on which automated leadership is based and people's limited experience with algorithms (Monteith & Glenn, 2016), this finding is surprising and contradicts the assumption that algorithms are often perceived as too complex and as generally not transparent (e.g., Glikson & Woolley, 2020). One explanation could be that the participants perceived human decisions as even less transparent. Participants may find it difficult to understand a person's decisions because not every supervisor shares the information that is used for decisions or reveals personal motives in a way that allows workers to accurately assess the underlying reasons for the supervisor's actions (Norman, Avolio, & Luthans, 2010). Algorithms, in contrast, follow specified rules (Lee, 2018; Merritt & Ilgen, 2008) that can be traced, and thus might have been perceived as more transparent.

Further, we found no support for our assumption regarding a difference in the perceived data processing capacity of human and automated leadership agents. It seems that the greater computing power and processing capacity of computers (Ferrás-Hernández, 2018; Jarrahi, 2018) was not perceived as a clear advantage over humans. Participants

may have ascribed similarly high levels of data processing capacity to humans because our scenarios contained decisions that were not considered complex enough to require the capacity to process large amounts of data. Another explanation could be that humans tend to generally overestimate their own abilities (Dunning, Meyerowitz, & Holzberg, 1989; Sundström, 2008) and are skeptical about the abilities of artificial intelligence (Hengstler, Enkel, & Duelli, 2016). In addition to the presented reasons, our study design may have promoted the effects for transparency and data processing capacity. We did not provide background information about the leadership agents and their decision-making processes. As a result, the participants potentially based their judgement on stereotypical assumptions or preexisting attitudes that may not involve differences between human and automated agents large enough for differences to be perceived.

Surprisingly, decision subject only moderated the relationship between integrity and trust, such that integrity was a stronger predictor for trust when the decision was disciplinary, i.e., a decision in which power differences were more salient. This corresponds with empirical findings in the domain of organizational trust that found a more pronounced relationship between integrity and trust in contexts with higher power differences between trustor and trustee but not for other trustworthiness factors (e.g., Colquitt et al., 2007; Pirson & Malhotra, 2011). Thus, our results suggest that the relationship of ability, benevolence, and transparency with trust are less dependent on power structures than the integrity–trust relationship.

Finally, additional analyses supported the influence of trust on workers' reactions to the leadership decision. In line with prior research and theory, trust was positively associated with the acceptance of the decision (Tost, 2011; Tyler, 1997), perceived fairness (Tyler, 1989, 1994), organizational attractiveness (Langer et al., 2018), and perceived organizational support (Tremblay et al., 2010). While these findings are not surprising in terms of the human leadership agent, it is important to note that our additional analyses indicated that also high trust in the automated leadership agent led to higher levels of the respective reaction. Accordingly, trust seems to be a key mechanism to influence the effectiveness of automated leadership.

5.2. Limitations and directions for future research

The findings of our study need to be considered in light of several limitations. First, experimental vignette methodology limits external validity and reduces the generalizability of our results because scenarios only provide an approximation of real-world experiences and thus are not capable of eliciting psychological reactions to actually experienced situations. However, the use of this method enabled us to test our hypotheses in a rather controlled setting that offers high internal validity and allows us to infer causality due to the manipulation of our independent variable (Robinson & Clore, 2001). Furthermore, our study represents one of the first examining trust in automated leadership, so experimental control is desirable to test our theory for the first time. Still, we followed best practice recommendations for vignette studies (Aguinis & Bradley, 2014) to ensure that participants perceived high scenario realism and imagined the situation as vivid and pictorial as possible. To this end, we provided participants with contextual background information about our scenarios and chose decisions that are likely to occur in natural settings. Nonetheless, our findings should be replicated in future studies that involve actual experiences with automated leadership. Thus, we strongly encourage scholars to examine our research question using other experimental approaches, e.g., a laboratory study in which participants perform a task guided by either an

automated or a human leadership agent. Moreover, field studies are needed with existing cases of automated leadership to replicate our findings and develop a deeper understanding of how people perceive and respond to this new form of leadership. In doing so, scholars should also aim for greater personal involvement of the participants in order to better reflect the various risk and power dynamics in the context of automated leadership. Further, it is desirable that future studies—conducted either in the laboratory or in the field—investigate how automated leadership relates to important leadership outcomes such as job performance and organizational citizenship behaviors. This would be an important next step towards a holistic understanding of automated leadership and its effectiveness.

Second, we included only two different decision subjects in our study. Both the allocation of monetary bonuses and promotions as well as the allocation of future vocational trainings represent decisions that are highly relevant for the worker, since they directly impact their career and life. However, given the huge variety of leadership decisions, it is desirable that future studies investigate different tasks and decision types to enable a generalization of our findings to other aspects of leadership. In doing so, researchers should take into account decisions that differ more clearly from each other (e.g., decisions that require rather mechanical/computational skills vs. decisions that require rather social/intuitive skills) in order to investigate their impact on the perceptions and reactions towards both the decisions and the leadership agents.

Third, the aim of the study was to understand the perceptions of algorithmic decisions when the automated leadership agent is presented as a “black box”. Accordingly, we did not provide specific details about its functioning to compare differences in assessments that result solely from knowing who the leadership agent is. Hence, our study focused on human versus automated agents without giving participants additional contextual information about e.g., the personality or competencies of the human leadership agent or specifics of the computer. Such context information about the human or the computer can, however, influence people’s perceptions and reactions. For instance, it is possible that an automated leadership agent may be perceived to perform better than a “bad” human agent, but not better than a “good” human agent. Similarly, context information or knowledge about algorithms could influence people’s perceptions and their responses towards automated leadership. Future research should thus explore how different attributes of automated or human leadership agents influence how leadership decisions are dealt with. In addition, studies should also consider different levels of knowledge about and experience with automated leadership (i.e., individual attributes of the worker). Even though our way of representing the algorithm was not ideal, we decided to focus on this approach to generate first information about trust in automated leadership.

Fourth, we did not counterbalance the order of vignette presentation in order to avoid our research model of becoming too complex due to another experimental factor (i.e., order). We have started with the assessment of human agents because humans are the current status quo of supervisors and decision agents. Thus, the following assessment of automated leadership agents should have been calibrated against the assessment of the current situation (Aguinis & Bradley, 2014). Although we are confident that order effects are not a major issue in our case, we do encourage future studies to replicate our findings in settings that systematically vary the order of assessments or follow between subject-designs. Furthermore, a replication of our study would be highly desirable to evaluate whether our findings can be confirmed in settings where workers remember past interactions or actually interact with automated leadership agents. As we refrained from explicitly testing our

experimental manipulations due to the clear and straight forward manipulation of both leadership agent and decision subject within the hypothetical scenarios presented in our vignettes, future studies might also employ manipulation checks when using our manipulations to ensure validity.

Finally, our study only captured the participants’ assessment of the respective leader in one specific situation. Accordingly, our results offer insights regarding people’s perception of automated versus human leadership agents at a specific point of time, but do not inform us about how perception and reaction unfold and change over time. Given that trust develops over time (e.g., Mayer et al., 1995), it is desirable that future research covers longer periods containing multiple decisions. Such studies will be especially valuable because possible differences between leadership agents might become clearer in repeated interactions.

5.3. Practical implications

Our study offers practical implications in the context of designing automated leadership and its application in organizations. First, our findings emphasize the importance of the leadership agent’s (whether human or computer) trustworthiness for the emergence of trust. Results further indicated the relevance of trust in the context of automated leadership as it was positively associated with the several outcome measures such as perceived fairness of the decision and perceived organizational support. Thus, the design of automated leadership agents should be human-centered to accommodate the needs, perceptions, and behaviors of a human user, thereby achieving high trustworthiness perceptions.

Second, the direct comparison of a computer versus a human leadership agent identified advantages and disadvantages that people perceive in both agents. These insights can help practitioners to decide which decisions and functions should be allocated to computers or not based on the required skills and qualities. It should be noted that many participants in our study attributed higher levels of integrity to computers, suggesting that automated leadership is considered more accurate and that algorithms could eliminate biases from leadership decisions (Monteith & Glenn, 2016). In the implementation of automated leadership, however, practitioners should be aware that algorithms are not infallible and that bias can sneak into algorithms in different ways. For instance, this may occur when algorithms make decisions based on training data that involve biased human decisions or reflect historical or social inequalities (Manyika, Silberg, & Presten, 2019). Hence, a careful consideration of our results both in the actual implementation and the communication of automated leadership to workers can assist practitioners in building efficient workplaces, but also in ensuring that workers can trust in and feel good about automated leadership.

5.4. Conclusion

Given the advancing prevalence of automated leadership, there is a need to understand how people would perceive and respond to it. The present study is one of the first addressing the question if workers would trust in automated decisions in the leadership context as compared to human decisions, thus advancing the knowledge in this new and fascinating field. Our results emphasize the importance of leadership agents’ perceived trustworthiness and the resulting trust in both human and computer agents, identifying trust as a key factor for the acceptance and the success of automated leadership.

Credit author statement

Miriam Höddinghaus: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization, Dominik Sondern: Methodology, Software, Validation, Formal analysis, Data curation, Writing – review & editing, Guido

Hertel: Conceptualization, Methodology, Writing – review & editing.

Declaration of competing interest

None.

Appendix A*Vignettes⁵**Instruction and Scenario*

In the following, we will describe an application scenario. Please read this carefully and take your time. Imagine that you are in the situation described. Try to put yourself in the position of the scenario and imagine the situation as vivid and pictorial as possible.

You are currently looking for a new job and have already completed the application phase. After some interviews with different companies, you now have two concrete job offers. Both companies, “Chemalux” and “AlphaChemics”, are operating in the chemical industry and would like to employ you in their respective quality assurance departments. The two offers hardly differ from each other. The positions correspond to your wishes and the basic salary as well as other basic conditions (vacation, benefits, etc.) are identical. For both companies, the share of variable remuneration is 60%. There are differences only with regard to the design of some decision-making processes. These differences relate to decisions that would directly affect you in your new position.

Vignette

Your *personal development/monetary pay for your job* is very important to you. Accordingly, you also attach great importance to the *personnel development programs/to possible career opportunities* offered by your potential employer.

In the quality assurance department at “**Chemalux**”/“**Alpha Chemics**”, the decisions regarding *the allocation and selection of annual workshops and training courses for each worker/the allocation and the amount of the monthly bonuses for each worker as well as promotion decisions* are made by **the head of department, Mr Schneider/a computer program**.

In order to decide *which activities will be necessary and meaningful next year/whether and to what extent bonuses are paid out and which worker is promoted*, **the head of department orients himself on/the computer program uses data on the existing competencies and vacancies within the company/the current performance of the workers as well as their medium and long-term development with regard to the achievement of set goals**. In this way, *workers are supported in a targeted and individual manner/can be remunerated and promoted on an individual and performance-related basis*.

Accordingly, the decision about *which annual training you will attend/your variable payment and promotions* will be made by **your department head, Mr. Schneider/a computer program**.

Appendix B

All items measured in the current study.

Construct	Item
Ability (data processing)	I think X has the competence to include all necessary information in the decision-making process.
	I think X is able to process all data necessary for reaching a decision.
	I think X is able to consider all necessary data when making a decision.
Ability (adaptability)	I think X can flexibly consider different circumstances when making decisions.
	I think X has the competence to adapt its decision to different circumstances.
	I think X is able to react flexibly to circumstances in the decision-making process.
Benevolence	I think X would put my interests first.
	I think X would always keep my interests in mind when making decisions.
	I think X would want to understand my needs and preferences regarding the decision.
Integrity	I think X makes unbiased decisions.
	I think X is honest and fair.
	I think X is incorruptible.
Transparency	I think I could understand the decision-making processes of X very well.
	I think I could see through X's decision-making process.
	I think the decision-making processes of X are clear and transparent.
Trust	I would heavily rely on X.
	I would trust X completely.
	I would feel comfortable relying on X.
Fairness	I think X makes fair decisions.
	I think the way X makes decisions is fair.
	I think the decisions of X are fair.
Acceptance	I think I would accept X's decision.
	I think I would agree with X's decision.

(continued on next page)

⁵ Vignettes translated to English, original in German. Manipulation of leadership agent in bold; manipulation of decision subject in italics

(continued)

Construct	Item
Org. attractiveness	I think I would endorse X's decision and act accordingly.
	I would like to work for <i>Chemalux/AlphaChemics</i> .
	I think <i>Chemalux/AlphaChemics</i> is attractive as a place for employment.
Org. support	I would accept a job offer from <i>Chemalux/AlphaChemics</i> .
	I think <i>Chemalux/AlphaChemics</i> would care about my career development.
	I think <i>Chemalux/AlphaChemics</i> would take an interest in my career.
Propensity to trust	I think <i>Chemalux/AlphaChemics</i> would take care of me financially.
	I think I would receive generous financial support from <i>Chemalux/AlphaChemics</i> .
	I think that most people basically have good intentions.
Trust in technology	I think that most people I deal with are honest and trustworthy.
	My first reaction is to trust people.
	I tend to assume the best of others.
Technology commitment	I have quite a lot of trust in human nature.
	I usually trust a technology until it gives me a reason not to trust it.
	I generally give a technology the benefit of the doubt when I first use it.
	My typical approach is to trust new technologies until they prove to me that I shouldn't trust them.
	I am very curious about new technical developments.
	For me, dealing with technical innovations mostly represents an excessive demand.
	I find it hard to deal with new technology—mostly I am just not able to do so.
	It is up to me whether I succeed in using new technical developments—this has little to do with chance or luck.
	I am always interested in using the latest technical devices.
	Dealing with modern technology, I am often afraid of failing.
	If I have difficulties in dealing with technology, it ultimately depends solely on me to solve them.
	If I had the opportunity, I would use technical products much more often than I do now.
	I am afraid that I rather destroy technological innovations than that I use them properly.
	What happens when I deal with new technical developments is ultimately under my control.
	I quickly take a liking to new technical developments.
Work values	Whether I am successful in using modern technology depends largely on me.
	How important to you is money (i.e., earning a lot of money) in your job?
	How important to you is career (i.e., advancing your career) in your job?
	How important to you is learning (i.e., continuously learning new skills and acquiring new knowledge) in your job?
	How important to you is leadership (i.e., having a fair and considerate supervisor) in your job?
	How important to you is autonomy (i.e., working independently and autonomously) in your job?
Perceived risk	How would you characterize the decision about the allocation of <i>monthly bonus payments/trainings and workshops</i> ? (significant opportunity/significant risk)
	How would you characterize the decision about the allocation of <i>monthly bonus payments/trainings and workshops</i> ? (high potential for gain/high potential for loss)
	How would you characterize the decision about the allocation of <i>monthly bonus payments/trainings and workshops</i> ? (very positive situation/very negative situation)
	What is the likelihood that you will be satisfied with the decision about the allocation of <i>monthly bonus payments/trainings and workshops</i> ? (very likely/very unlikely)

Note. Items translated from German. Depending on the respective vignette, X was replaced by either *Mr Schneider* or *the computer*.

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